

Question	Marking Guidance	Mark	Comments		
06.1	1. 1 long and 1 short chromosome, each made up of 2 chromatids held (by centromere), in each cell of 1st division; 2. 1 long and 1 short (separate) chromosome in each cell of 2nd division;	2	2. Allow ECF for correct chromosomes shown in each cell from candidate's 1st division cells. 2. Ignore drawing of centromere.		
06.2	<table><tr><td>52</td><td>4</td></tr></table> ;; Allow 1 mark for numbers totalling 56 except 14/42 - repetition of observed values. If table is blank, award 1 mark for evidence of 56.	52	4	2	Both 52 and 4 required in table for two marks, do not credit 52 or 4 for one mark. Award 1 max for answers not given as whole numbers.
52	4				
06.3	1. There is a less than 0.05/5% <u>probability</u> that the <u>difference</u>(s) (between observed and expected) occurred by <u>chance</u>; 2. Calculated value is greater than critical value so the null hypothesis can be rejected; 3. (The scientists can conclude that) the proportion of plants that produce 2n gametes does change from one breeding cycle to the next;	2 max	1. Reject 'results (without reference to difference) occurring by chance'. Overall max 1 with this statement. 1. Accept 'there is a greater than 0.95/95% <u>probability</u> that the difference did not occur by <u>chance</u>'. 1. and 2. Ignore 'difference is significant' 2. Do not accept 'P value' for 'critical value'.		

Question	Marking Guidance	Mark	Comments
06.4	1. The scientists selected/used for breeding plants that produced 2n gametes; 2. (So these plants) passed on their alleles (for production of 2n gametes to the next generation); 3. The frequency of alleles for production of 2n gametes increased (in the population).	3	For 'production of 2n gametes' accept 'abnormal meiosis'. 1. Answer must be in context of the scientists selecting plants to breed. Accept 'artificial selection' or 'selectively bred'. 2 and 3. Both mark points can be awarded if one correct reference is made to alleles (in either context). 3. Do not accept 'number' for frequency. Accept converse answers linked to plants that produce n gametes.

